
IFIP Transactions A: Computer Science and Technology



International Federation for Information Processing

Technical Committees:

Software: Theory and Practice (TC2)
Education (TC3)
System Modelling and Optimization (TC7)
Information Systems (TC8)
Relationship between Computers and Society (TC9)
Computer Systems Technology (TC10)
Security and Protection in Information Processing Systems (TC11)
Artificial Intelligence (TC12)
Human-Computer Interaction (TC13)
Foundations of Computer Science (SG14)

IFIP Transactions Editorial Policy Board

The IFIP Transactions Editorial Policy Board is responsible for the overall scientific quality of the IFIP Transactions through a stringent review and selection process.

Chairman

G.J. Morris, UK

Members

D. Khakhar, Sweden

Lee Poh Aun, Malaysia

M. Tienari, Finland

P.C. Poole (TC2)

P. Bollerslev (TC3)

T. Mikami (TC5)

O. Spaniol (TC6)

P. Thoft-Christensen (TC7)

G.B. Davis (TC8)

K. Brunnstein (TC9)

E. Hörbst (TC10)

W.J. Caelli (TC11)

R. Meersman (TC12)

B. Shackel (TC13)

J. Gruska (SG14)

IFIP Transactions Abstracted/Indexed in:

INSPEC Information Services

Index to Scientific & Technical Proceedings®

CompuMath Citation Index®, Research Alert™, and SciSearch®

COMPUTER SECURITY

Proceedings of the IFIP TC11 Ninth International Conference on
Information Security, IFIP/Sec'93
Toronto, Canada, 12-14 May, 1993

Edited by

E. GRAHAM DOUGALL

ManuLife Financial
Toronto, Ontario, Canada



1993

NORTH-HOLLAND
AMSTERDAM • LONDON • NEW YORK • TOKYO

ELSEVIER SCIENCE PUBLISHERS B.V.
Sara Burgerhartstraat 25
P.O. Box 211, 1000 AE Amsterdam, The Netherlands

Library of Congress Cataloging-in-Publication Data

IFIP TC11 International Conference on Information Security (9th : 1993
: Toronto, Ont.)

Computer security : proceedings of the IFIP TC11 Ninth
International Conference on Information Security, Toronto, Canada,
12-14 May, 1993 / edited by E. Graham Dougall.

p. cm. -- (IFIP transactions. A, Computer science and
technology : A-37)

Includes bibliographical references.

ISBN 0-444-81748-4

1. Computer security--Congresses. I. Dougall, E. Graham.

II. Title. III. Series.

QA76.9.A25I45 1993

005.8--dc20

93-34965

CIP

Keywords are chosen from the ACM Computing Reviews Classification System, ©1991, with permission.
Details of the full classification system are available from
ACM, 11 West 42nd St., New York, NY 10036, USA.

ISBN: 0 444 81748 4

ISSN: 0926-5473

© 1993 IFIP. All rights reserved.

No part of this publication may be reproduced, stored in a retrieval system or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without the prior written permission of the publisher, Elsevier Science Publishers B.V., Copyright & Permissions Department, P.O. Box 521, 1000 AM Amsterdam, The Netherlands.

Special regulations for readers in the U.S.A. - This publication has been registered with the Copyright Clearance Center Inc. (CCC), Salem, Massachusetts. Information can be obtained from the CCC about conditions under which photocopies of parts of this publication may be made in the U.S.A. All other copyright questions, including photocopying outside of the U.S.A., should be referred to the publisher, Elsevier Science Publishers B.V., unless otherwise specified.

No responsibility is assumed by the publisher or by IFIP for any injury and/or damage to persons or property as a matter of products liability, negligence or otherwise, or from any use or operation of any methods, products, instructions or ideas contained in the material herein.

pp. 1-12, 13-22, 23-30, 195-206, 287-300, 313-326, 359-366: Copyright not transferred

This book is printed on acid-free paper.

Printed in The Netherlands

Preface

This volume contains the papers presented at IFIP/Sec'93 at Deerhurst Inn, Huntsville, Ontario, Canada from May 12 to 14 1993. This conference was second one to be hosted by Canada, the first was IFIP/Sec'84. As I was involved in the 84 conference it was a pleasure renewing old friends and forming new ones.

The theme of the conference was *Discovering Tomorrow*. We were fortunate in having Patrick Gallagher, the Director of the United States Government's National Computer Security Center which is part of the National Security Agency as the opening keynote speaker. He addressed the changing outlook of his Center.

IFIP/Sec'93 was honoured in having the first presentation of the Kristian Beckman Award. This award recognizes the contribution Kristian made in the organization of the first conference in Stockholm in 1983 and subsequent founding of TC11. It will be presented annually at the TC11 conferences to an individual who has made an outstanding contribution to computer security internationally. The acceptance speech of this year's recipient, Dr. Harold J. Highland is included in this volume.

I would like to thank all those involved in making IFIP/Sec'93 a success and look forward to meeting in Aruba in 1994.

E. Graham Dougall

IFIP/Sec'93

IFIP/Sec'93 was organized under the auspices of IFIP TC 11 by the National Security SIG of the Canadian Information Processing Society (CIPS) and the Toronto Chapter of the Information Systems Security Association (ISSA).

Organizing Committee

David Batchelor, Chair
(Canada)
Dan Eng
David Gamey
Tim Haist
Beth Kendall
Peter Kingston
June S. Morrison

Program Committee

Graham Dougall, Co Chair

Jim Finch, Co Chair (Canada)
John Beatson, (New Zealand)
Bertil Fortie (Netherlands)
Darren Jones (Canada)
Rick Koenig (USA)
Hal Tipton (USA)

Sponsors

IFIP/Sec'93 would like to thank the following organizations for their support..

Computer Associates
Concord - Eracom Computer Ltd.
The Kingston Group
Nabisco Brands Ltd.
Sussex Systems Ltd.

IFIP/Sec'93 Table of Contents

Kristian Beckman Award Address	1
A View of Information Security Tomorrow Harold Joseph Highland	
Opening Keynote Address	13
The Evolution of IT Security Convergence Patrick R. Gallagher, Jr.	
Teaching security basics: The Importance of When and How	23
Viiveke Fåk and Amund Hunstad	
The MSc in Information Security at Royal Holloway	31
Dieter Gollmann	
Information Security in the Small Systems Context: A Framework for Understanding	37
Guy G. Gable and Harold Joseph Highland	
Developing Awareness, Training and Education: A Cost Effective Tool for Maintaining System Integrity	53
Corey D. Schou, W. Vic Maconachy and James Frost	
Teaching Computer Security	65
Matt Bishop	
The Filter Model of Information Security: A Conceptual Model for Education and Training	75
A. R. Smith	
Shamir's Scheme Says It all	91
Ed Dawson and Diane Donovan	
LUC: A New Public Key System	103
Peter J. Smith and Michael J.J. Lennon	
The Encapsulated Security Services Interface (ESSI)	119
Ping Lin	

Context-Dependent Access Control in Distributed Systems Hermann Strack and Kwok-Yan Lam	137
Internal Control by Objectives: The Functional Control Matrix Karl H. Krueger	157
International Standards and Organizational Security Needs: Bridging the Gap C.J. Bosch, J.H.P. Eloff and J.M. Carroll	171
The Risk Data Repository: a Novel Approach to Security Risk Modelling A. M. Anderson, D. Longley and A.B. Tickle	185
Turning Multiple Evaluated Products into Trusted Systems with Assurance D. Gambel and J. Fowler	195
Historical Labels in Open Distributed IT Systems: An ITSEC/ECMA Specification Stewart Kowalski	207
A Model for Organising Information Security Documentation L.M. du Toit and S.H. von Solms	227
Planning For The EDI of Tomorrow Using Electronic Document Authorization S. Russell	243
The EDP Auditor: Disappearing Or Adapting Guy. G. Gable and Gordon B. Davis	253
Electronic Data Interface(EDI) Security and Audit: a Practical Approach Ivan Ekebrink	267
A Generic Security Platform for Workstations Björn Lindberg	277
Evaluation of an Academic Programme in IT Security 1985-1990 Louise Yngström	287
Fighting The Viruses: Practical Anti-Virus Training Mikael Larsson	301

A Quantitative Accreditation Model - Assessing the Value of Assurance M. Ohlin	313
Superseding Manual Generation of Access Control Specification - From Policies to Profiles D. Pottas and S.H. von Solms	327
Extending the ISO Access Framework for Multiple Policies Marshall D. Abrams and Michael V. Joyce	343
Evolution of IT Security in India - Implementation Issues K. Subramanian	359
Legal Response to a Computer Crime - Retrospect of a "Mere Chance" Case Juhani Saari	369
Information Systems Security and Fraud Prevention in Office Automation Systems Philip M. Stanley	375
A Process Approach to Information Security Management R. von Solms, S.H. von Solms and J.M. Carroll	385
MRA: A Computational Technique for Security in High-Performance Systems Mahdi Abdelguerfi, Andrea Dunham and Wayne Patterson	401